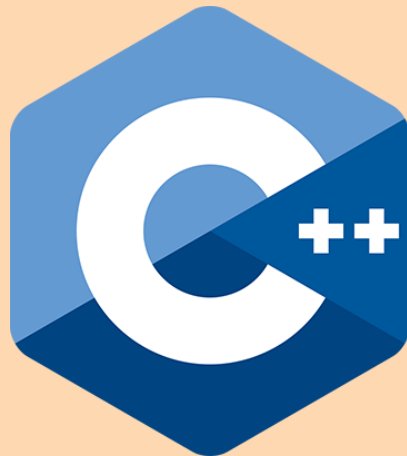


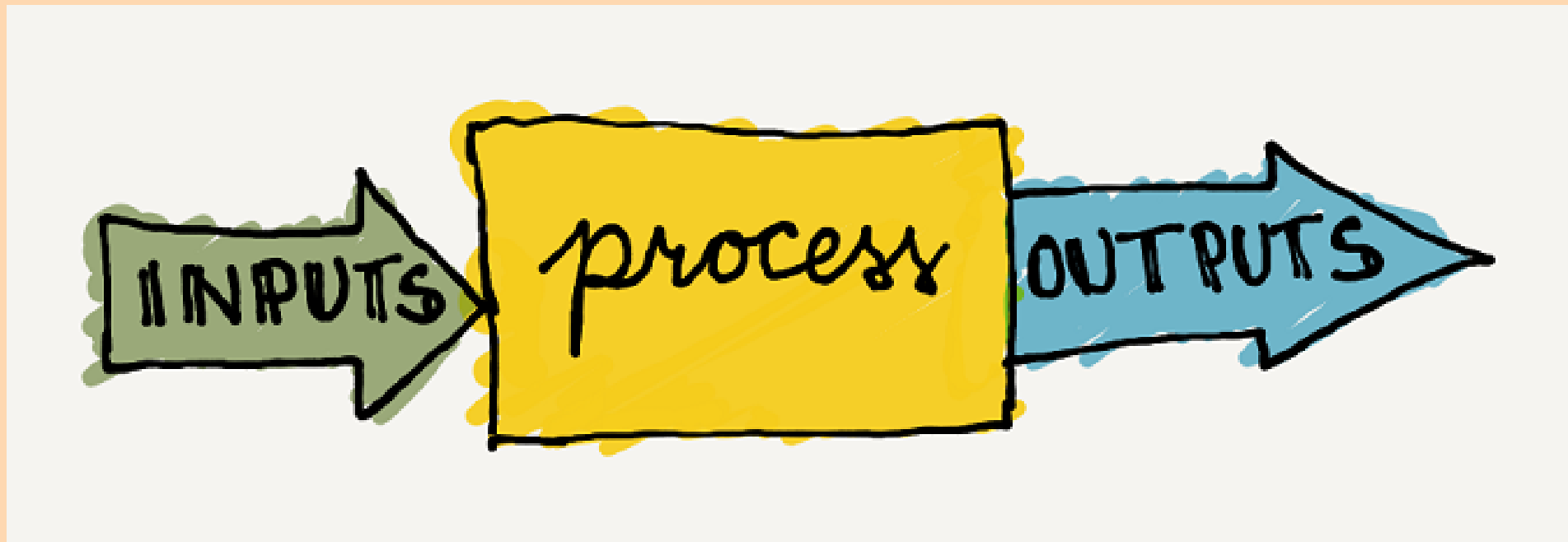
# 1. VARIABLES, TYPES, I/O

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# Overview

**Program** = **Input** + **Data** + **Operations** + **Output**



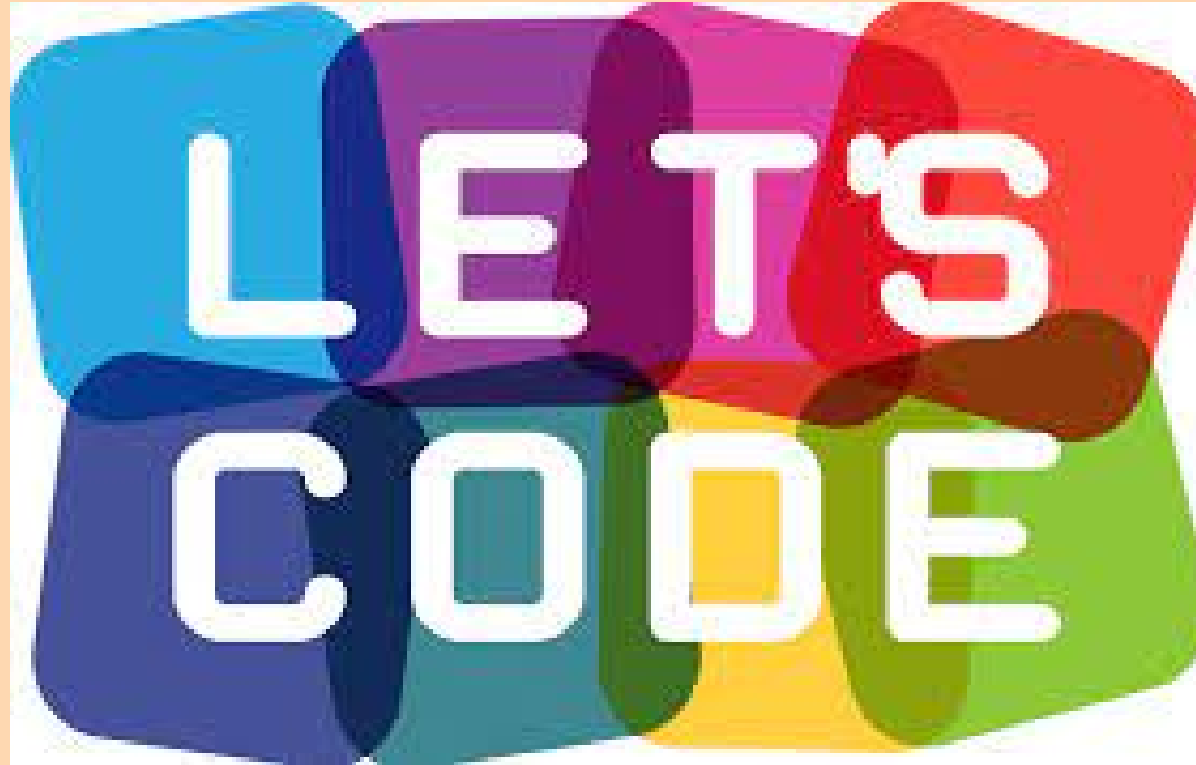
# Overview

**Variables (变量)      Data Types (数据类型)**

**Operators (运算符)      Expressions (表达式)**

**Assignment Statements (赋值语句)**

**I/O (Input/Output; 输入输出)**



**circle.cpp   sphere.cpp**

**admin.cpp   admin-cin.cpp**

# Circle

Given a **radius** (say 10) of a circle,  
to compute its **circumference** and **area**.

$$L = 2\pi r \quad S = \pi r^2$$

- 每个结果各占一行
- 小数点后保留两位

# Declaration/Definition (声明/定义)

```
int radius{10};
```

- Declare/Define a *variable* called `radius`.
- `radius` refers to a *location* in memory.
- The **type** of `radius` is `int` (integer).
- `radius` is *initialized* to 10.
- You can *assign* other integers to `radius`.

# Type is a kind of Abstraction



Type: set of allowed **values & operations**

# Identifiers (标识符)

```
int radius{10};
```

`radius` is an *identifier*.

**Warning:** Do *not* start with `_`, which are reserved by C++.

**Always use `meaningful` identifiers in a `uniform` style!!!**



```
int radius;
```

```
cout << radius;
```

Undefined Behavior (UB)

# Operators & Expressions

```
double circumference{2 * PI * radius};
```

# Sphere

Given a **radius** (say 100) of a sphere,  
to compute its **surface area** and **volume**.

$$A = 4\pi r^2 \quad V = \frac{4}{3}\pi r^3$$

- 每个结果占 1 行
- 小数点后保留 4 位
- 每个结果至少占 15 字符, 左对齐
  - `_____ : surface_area`
  - `_____ : volume`

# Input/output manipulators (操纵符)

```
#include <iomanip>
```

<https://en.cppreference.com/cpp/io/manip>

**std::format**

**<https://en.cppreference.com/cpp/utility/format/spec>**

# Format Specification

```
{:[align][width][.precision][type]}
```

- **d**: decimal `int`
- **f**: `double`
- **S**: `string` (not necessarily)
- **C**: `char` (not necessarily)

```
{:[align][width][.precision][type]}
```

- <: left-justified
- >: right-justified
- ^: centered

```
{:[align][width][.precision][type]}
```

- minimum field width
- padded with spaces if it has fewer characters

```
{:[align][width][.precision][type]}
```

- **f**: number of digits after **.**
- **s**: maximum number of characters

# A (Naive) Administration System

- Name (EN)
- Gender (F/M)
- Birthday (mm-dd-yyyy)
- Weekday (Xyz.)
- C++
- Music
- Medicine
- Mean (.d)
- Standard Deviation (.dd)
- Ranking (%)

未來的主人翁



羅大佑

**For 罗大佑 only:**

- 每组信息占一行
- 各项信息使用 `\t` 间隔
- 各项信息遵循特定格式要求

# char

|           | _0          | _1          | _2          | _3          | _4          | _5          | _6          | _7          | _8          | _9         | _A          | _B          | _C         | _D         | _E         | _F          |
|-----------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|------------|-------------|-------------|------------|------------|------------|-------------|
| 0_<br>0   | NUL<br>0000 | SOH<br>0001 | STX<br>0002 | ETX<br>0003 | EOT<br>0004 | ENQ<br>0005 | ACK<br>0006 | BEL<br>0007 | BS<br>0008  | HT<br>0009 | LF<br>000A  | VT<br>000B  | FF<br>000C | CR<br>000D | SO<br>000E | SI<br>000F  |
| 1_<br>16  | DLE<br>0010 | DC1<br>0011 | DC2<br>0012 | DC3<br>0013 | DC4<br>0014 | NAK<br>0015 | SYN<br>0016 | ETB<br>0017 | CAN<br>0018 | EM<br>0019 | SUB<br>001A | ESC<br>001B | FS<br>001C | GS<br>001D | RS<br>001E | US<br>001F  |
| 2_<br>32  | SP<br>0020  | !<br>0021   | "<br>0022   | #<br>0023   | \$<br>0024  | %<br>0025   | &<br>0026   | '<br>0027   | (<br>0028   | )<br>0029  | *<br>002A   | +<br>002B   | ,<br>002C  | -<br>002D  | .<br>002E  | /<br>002F   |
| 3_<br>48  | 0<br>0030   | 1<br>0031   | 2<br>0032   | 3<br>0033   | 4<br>0034   | 5<br>0035   | 6<br>0036   | 7<br>0037   | 8<br>0038   | 9<br>0039  | :<br>003A   | ;<br>003B   | <<br>003C  | =<br>003D  | ><br>003E  | ?<br>003F   |
| 4_<br>64  | @<br>0040   | A<br>0041   | B<br>0042   | C<br>0043   | D<br>0044   | E<br>0045   | F<br>0046   | G<br>0047   | H<br>0048   | I<br>0049  | J<br>004A   | K<br>004B   | L<br>004C  | M<br>004D  | N<br>004E  | O<br>004F   |
| 5_<br>80  | P<br>0050   | Q<br>0051   | R<br>0052   | S<br>0053   | T<br>0054   | U<br>0055   | V<br>0056   | W<br>0057   | X<br>0058   | Y<br>0059  | Z<br>005A   | [<br>005B   | \<br>005C  | ]<br>005D  | ^<br>005E  | _<br>005F   |
| 6_<br>96  | `<br>0060   | a<br>0061   | b<br>0062   | c<br>0063   | d<br>0064   | e<br>0065   | f<br>0066   | g<br>0067   | h<br>0068   | i<br>0069  | j<br>006A   | k<br>006B   | l<br>006C  | m<br>006D  | n<br>006E  | o<br>006F   |
| 7_<br>112 | p<br>0070   | q<br>0071   | r<br>0072   | s<br>0073   | t<br>0074   | u<br>0075   | v<br>0076   | w<br>0077   | x<br>0078   | y<br>0079  | z<br>007A   | {<br>007B   | <br>007C   | }<br>007D  | ~<br>007E  | DEL<br>007F |

A `char` is actually an `int`.

# C++ String

```
std::string first_name{"Tayu"};
```



未来的主人翁

# Review

**Program** = **Input** + **Data** + **Operations** + **Output**

